
FWAT Documentation

Release v1.1

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This is the documentation for the fortran inversion package of **FWAT**, which provides an inversion workflow of Full Waveform Adjoint Tomography for seismic wavefield imaging based on the **SPECFEM3D**. For further information and contact information please see below website:

- Gitlab repository of *FWAT*: https://github.com/wangkaim8/fwat_release.

If you use FWAT for your research and prepare publications, please consider citing **FWAT**:

- Wang, K., Yang, Y., Basini, P., Tong P., Tape, C. and Liu Q., 2018. Refined crustal and uppermost structure of southern California by ambient noise adjoint tomography. *Geophysical Journal International*, 215(3), 844-863.
- Wang, K., Yang, Y., Jiang, C., Wang, Y., Tong, P., Liu, T. and Liu, Q., 2021. Adjoint tomography of ambient noise data and teleseismic P waves: Methodology and applications to central California. *Journal of Geophysical Research: Solid Earth*, 126(6), p.e2021JB021648.

1. INTRODUCTION

The basic idea of **FWAT** is to iteratively minimize the misfit function of measurements between data and synthetics. According to the job submission schedule, the inversion can be divided into three stages as shown in Figure 1. In **Stage I**, both forward and adjoint simulations are conducted in one PBS/SLURM job (including misfit measurement) to obtain the event kernel of each virtual source (or master station). After all the simulations are finished, event kernels are summed, preconditioned, and smoothed in **Stage II** to obtain the final misfit gradient. In **Stage III**, a line search method is adopted to determine the optimal step length for model updating.

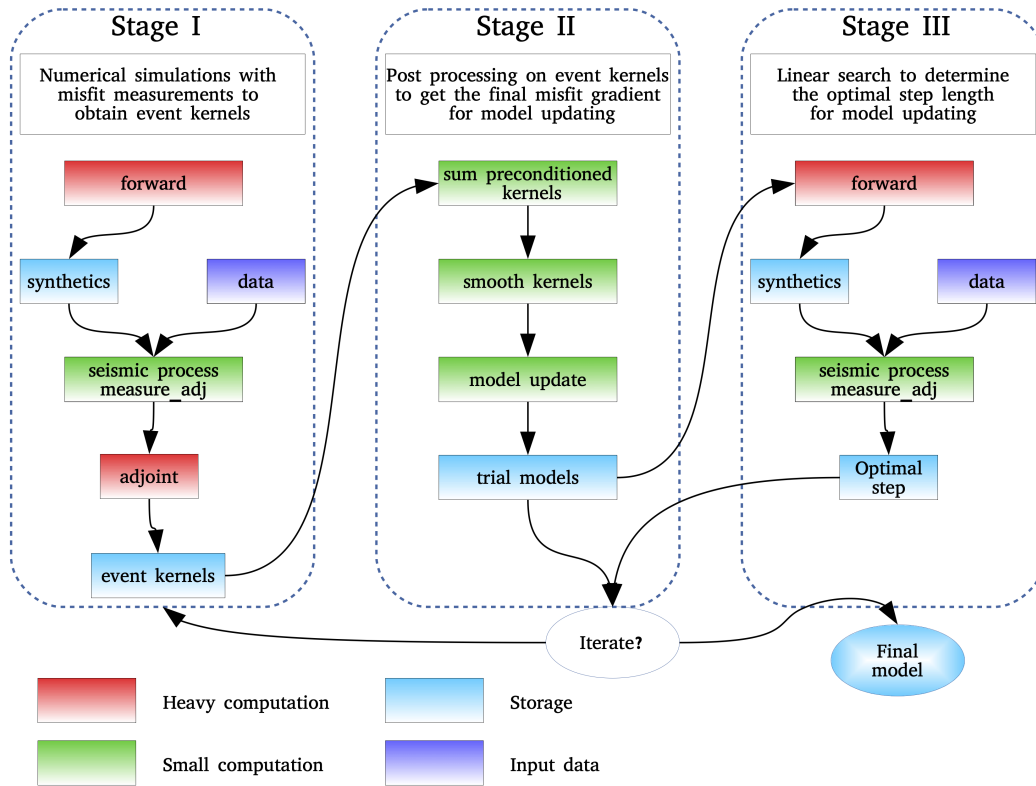


Fig. 1: Figure 1. Workflow of FWAT.

FWAT will generate several main programs in the **bin** directory:

- *xfwat0_forward_data* is a forward only MPI program, normally used for synthetic tests.
- *xfwat1_fwd_measure_adj* is a MPI program for **Stage I**
- *xfwat2_postproc_opt* is a MPI program for **Stage II**
- *xfwat3_linesearch* is a MPI program for **Stage III**
- *xfullwave_adjoint_tomo* is not finished yet and planned to combine *xfwat1*, *xfwat2* and *xfwat3* for fast computations of small-scale inversions.
- *xmeasure_adj* is a single program to test the [measure_adj](#)
- *xflexwin* is a single program to test the [FLEXWIN](#)

1.1 Support

Kai Wang is currently supported by the University of Science and Technology of China (USTC) Startup Fund (2023/2-present).

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1.2 Future update plan

There are several aspects can be considered for improvements in the future, including:

1. Add an inversion option for local earthquake (leq) tomography ([ngoing](#), Kai Wang).
2. Add CMT3D for source inversions prior to structure inversions ([ngoing](#), Kai Wang).
3. Python tool for model visualization ([ongoing](#), Mijian Xu).
4. Add multicomponent ANAT ([ngoing](#), Kai Wang).
5. Update **optimize** to invert for azimuthal anisotropy ([ngoing](#), Kai Wang).

2. INSTALL THE PACKAGE

2.1 Install FWAT

The latest release of FWAT is FWAT v1.1, which is based on the stable version of [SPECFEM3D](#) last modified on Aug 30, 2018 with a commit number of 42abac6d18 on Github.

```
$: git clone --recursive https://github.com/geodynamics/specfem3d.git
$: cd specfem3d
$: git checkout -b fwat 42abac6d18
$: cd src
$: git clone --recursive https://gitlab.com/wangkaim8/fwat.git
$: mv fwat fullwave_adjoint_tomo
$: cd ..

==> Change setup/constants_tomography.h.in
logical, parameter :: USE_ALPHA_BETA_RHO = .true.
logical, parameter :: USE_ALPHA_BETA_RHO_TISO = .false. ! Kai added
==> Change src/shared/shared_par.F90
logical :: ANISOTROPIC_KL,SAVE_TRANSVERSE_KL,APPROXIMATE_HESS_KL,SAVE_MOHO_MESH
logical :: SAVE_AZIMUTH_KL ! Kai added

==> Install FLEXWIN ttimes library
$: cd src/fullwave_adjoint_tomo/preproc_flexwin/ttimes_mod
$: make -f make_gfortran
$: make -f make_gfortran install
##### Install with intel ifort #####
$: ./configure FC=ifort CC=icc
==> Change Makefile:
1. set MPILIBS = -Lsrc/fullwave_adjoint_tomo/preproc_flexwin/ttimes_mod -ltau -lm -lmkl
##### Install with GNU gfortran #####
First,install BLAS, LAPACK
In Ubuntu, run
$: sudo apt-get install libblas-dev liblapack-dev
In linux clusters, try to module load lapack
In Mac OS, compile from source codes.
Second, install fwat with specfem3d by
$: ./configure FC=gfortran CC=gcc MPIFC=mpif90 --with-mpi
==> Change Makefile:
1. set MPILIBS = -Lsrc/fullwave_adjoint_tomo/preproc_flexwin/ttimes_mod -ltau -lm -
  ↳llapack -lblas
FLAGS_CHECK = -std=gnu -fimplicit-none -frange-check -fmax-errors=10 -pedantic -pedantic-
```

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```

↪errors
-Waliasing -Wampersand -Wcharacter-truncation -Wline-truncation -Wsurprising -Wno-tabs -
↪Wunderflow
-ffpe-trap=invalid,zero,overflow -Wunused -O3 -finline-functions
#####

2. replace
SUBDIRS = \
    tomography \
    tomography/postprocess_sensitivity_kernels \
with
SUBDIRS = \
    fullwave_adjoint_tomo/optimization \
    fullwave_adjoint_tomo \
    fullwave_adjoint_tomo/post_processing \

3. add
DEFAULT = \
    fwat_postproc \
    optimization \
    fullwave_adjoint_tomo \
delete
all: postprocess tomography
add:
all: fwat_postproc optimization fullwave_adjoint_tomo

```

2.2 Notes on segmentation fault/floating

You may need to change some parameters to reduce the memory requirement to avoid

↪ "segmentation fault" or set them larger to avoid "floating point exception"

====> Change fullwave_adjoint_tomo_par.f90

LNPT=17

NDIM = 10000000

NWINDOWS = 2000

In Mac OS, if NDIM is too large, you may get error "error: unexpected token in '.section

↪ ' directive

.section .data.rel.ro.local"

2.3 How to Modify FWAT to fit your version of SPECFEM3D

If you have to use a different version, you may take the following steps to modify the **FWAT** package in order to fit your version of *SPECFEM3D*.

```

step1 : Modify setup_sources_receivers_fwat.f90 and save_adjoint_kernels_fwat.f90
a) The file setup_sources_receivers_fwat.f90 is modified from the original file
    src/specfem3d/setup_sources_receivers.f90 to use a user-defined a source_fname
    and station_fname instead of the original CMTSOLUTION/FORCESOLUTION and STATIONS.

```

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You can modify this file according to your version of
 src/specfem3d/setup_sources_receivers.f90

b) Similarly, modify save_adjoint_kernels_fwat.f90 according to your version of
 src/specfem3d/save_adjoint_kernels.f90

step 2: Change rule.mk **if** necessary. FWAT almost call all objects (*.o) that are
 required by the program "xspecfem3D". So refer to src/specfem3D/rule.mk **if**
 missing some *.o.

step 3: Copy ../tomography/postprocess_sensitivity_kernels/* to post_processing/.
 Discard the changes **in** optimization/rule.mk

step 4: Copy some files **in** optimization from ../tomography/ that read "proc*external_
 ↪ mesh.bin"

==> optimization/compute_kernel_integral.f90 (Change topo to trim(INPUT_
 ↪ DATABASES_DIR)

==> optimization/save_external_bin_m_up.f90

==> optimization/read_model.f90 (add irregular mesh)

Modify this file:

==> optimization/get_lbfgs_direction.f90 (This one should be carefully tested
 after modification)

Note FWAT uses some old programs **in** tomography to update
 the model and also add some additional programs:

xadd_model_iso_cg
 xadd_model_iso_lbfgs
 xadd_model_tiso_cg

!!! Check all codes that read and write "proc*external_mesh.bin"

Please report issues on the github page or contact the developers.

3. PACKAGE STRUCTURE

The **FWAT** package consists of some workflow-control parameter files and bash scripts in several folders. The structure of FWAT is summarized in Figure 2.

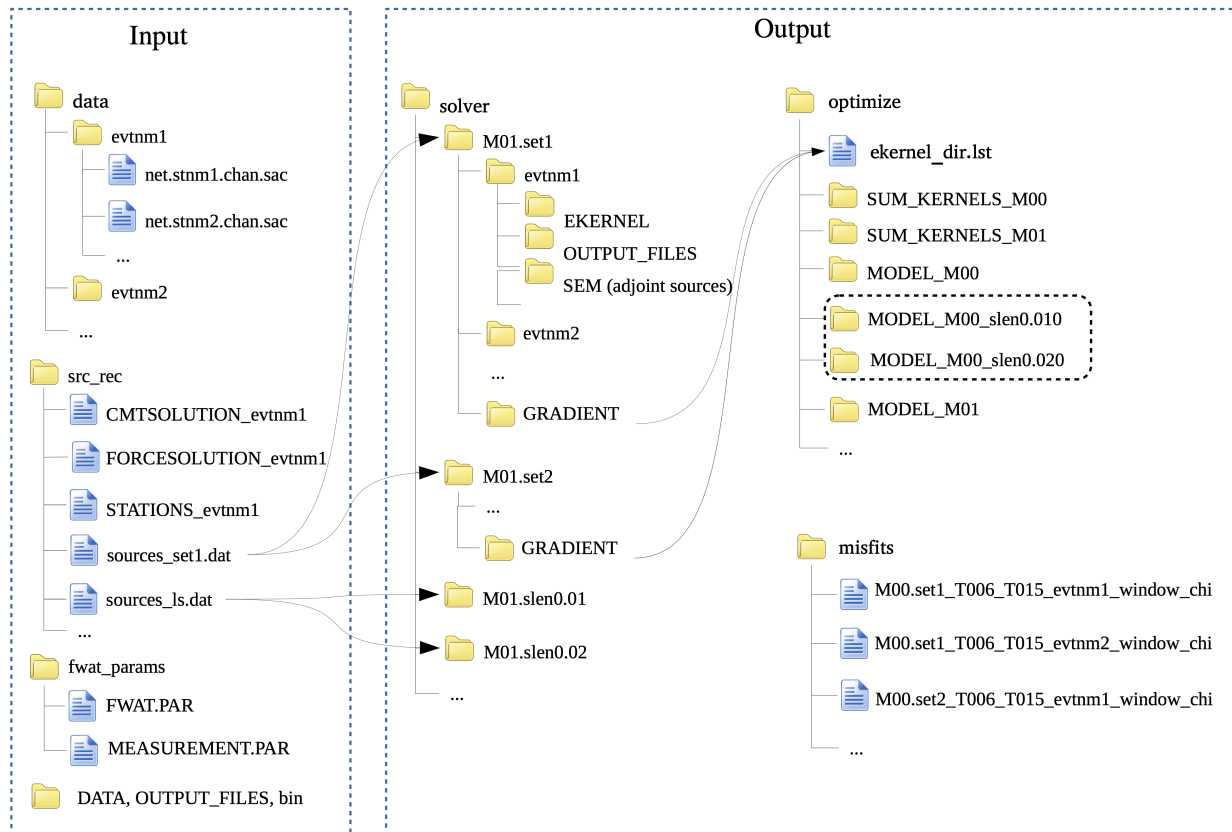


Fig. 1: Figure 2. Structure of FWAT.

3.1 Terminology

- **net**: network name, such as CI, TA, ...
- **stnm**: station name, such as CC01, CC02, ...
- **chan**: channel name, such as BXZ, ...
- **evtnm**: event name
- **slen**: step length

3.2 Inputs

- **data** contains sub-directories of seismic waveforms stored by event names. All the data are in SAC format and named as `./data/evtnm/net.sta.chan.sac`, where `evtnm` and `net` are event and network names, `sta` is station, `chan` is the channel. For example, `./data/CI.STC_Z/CI.BAK.BXZ.sac` is the EGF between a vertical virtual source CI.STC and a general station CI.BAK for vertical components. `./data/P1/T0.CC01.BXZ.sac` is the vertical component waveform recorded by station TO.CC01 from a teleseismic event named P1.

Note: CCFs requires sac headers: `knetwk kstnm kcmpnm dist az baz stlo stla evlo evla`. Remember to remove zero/NaN traces because it will cause errors when applying normalization to CCFs.

Use the script `rewrite_sac_header.bash` to rewrite sac headers and remove zero/NaN traces.

- **src_rec** stores all source and station information needed for the inversion. There are generally three types of files prepared for the *SPECFEM3D Cartesian*, namely `CMTSOLUTION_evtnm`, `FORCESOLUTION_evtnm`, and `STATIONS_evtnm`. The format of these files should be the same as described in the user manual of *SPECFEM3D Cartesian*. The other two files called by **FWAT** are `sources_set?.dat` and `sources_ls.dat`. `sources_set?.dat` is the name list of events (or virtual sources), which are used for making directories for simulations by event name. We can divide the events into different sets, such as `set1`, `set2`, ..., etc. `sources_ls.dat` contains the event names for line searches. The format of this two source list is:

```
event name | latitude | lon | elevation | bury depth
BK.KCC_Z 37.323600 -119.318700 888.000000 0.0
BK.PACP_Z 37.008000 -121.287000 844.000000 0.0
BK.PKD_Z 35.945200 -120.541600 583.000000 0.0
CI.ARV_Z 35.126900 -118.830100 258.000000 0.0
CI.BAK_Z 35.344400 -119.104400 116.000000 0.0
CI.CAR_Z 35.308200 -119.845800 765.000000 0.0
```

- **DATA, OUTPUT_FILES, bin**. These three inputs are essential for launching numerical simulations for *SPECFEM3D Cartesian*. Please refer to the user manual of *SPECFEM3D Cartesian* for the details.
- **fwat_params** contains input parameters for **FWAT**. There are two files including `FWAT.PAR` and `MEASUREMENT.PAR`.

3.3 Outputs

- **misfits** stores all the misfits files for a specific model. For example, all misfits of event set1 for model M01 are stored as `./misfits/M01.set1_T006_T015_evtnm1_window_chi`.
- **optimize** is where postprocessing procedure applied. All event kernels are processed in the directories `SUM_KERNELS_M??`, and model updating files are stored in `MODEL_M??`.
- **solver** is where numerical simulations are performed and stored. For `evtnm1` in the source list `set1`, the forward simulation root directory is `./solver/M01.set1/evtnm1`. For line search at step length of 0.010, the corresponding directory is `./solver/M01.slen0.010/evtnm1`.

3.4 Example scripts

3.4.1 Driven scripts

- `submit_job_fwat1.sh` is the driven script for running forward and adjoint simulations in **Stage I**. This script will call the PBS script `pbs_fwat1_fwd_measure_adj.sh` or SLURM script `sbash_fwat1_fwd_measure_adj.sh` and the results are stored in the directories **solver** and **misfit**.
- `submit_job_fwat2.sh` is the driven script to perform postprocessing of kernels and optimization for model update in **Stage II**. It will call the PBS script `pbs_fwat2_postproc_opt.sh` or SLURM script `sbash_fwat2_postproc_opt.sh` and the results are stored in the directories **optimize**.
- `submit_job_fwat3.sh` is the driven script to do forward simulations for a linear search in **Stage III**. This script will call the PBS script `pbs_fwat3_linesearch.sh` or SLURM script `sbash_fwat1_linesearch.sh` and the results are stored in the directories **solver** and **misfits**.

3.4.2 fwat_params

Examples of *FWAT.PAR* and *MEASUREMENT.PAR*

3.4.3 Plot

- **plots** contains some example scripts for plotting misfit, model, kernel, waveform.

4. TUTORIAL

This tutorial will teach you how to perform an iterative full waveform inversion with **FWAT** and **SPECFEM3D**.

- *1.1 Prepare inputs for FWAT*
- *1.2 Prepare inputs for SPECFEM3D and build mesh*
- *1.3 Prepare `fwat_params`*
- *1.4 Running forward and adjoint simulations (Stage I)*
- *1.5 Post-processing and model update (Stage II)*
- *1.6 Line search for optimal step length (Stage III)*

4.1 1.1 Prepare inputs for FWAT

Before running FWAT, the user should first make a new project by copy the folder **example_scripts**. For example, to build a new seismic project named *FWAT_test*, run the following commands:

```
$: mkdir FWAT_test
$: cd FWAT_test
$: cp -r ${path_to_specfem3d}/src/fullwave_adjoint_tomo/example_scripts/* .
```

After that, the user should write their own scripts to make **data** and **src_rec** in the format described in section 3. The corresponding example scripts are *generate_data.bash* and *mk_forcesolution.bash*.

4.2 1.2 Prepare inputs for SPECFEM3D and build mesh

Run the following commands for the preparation of *SPECFEM3D*:

```
$: ln -s ${path_to_specfem3d}/DATA .
$: ln -s ${path_to_specfem3d}/bin .
$: mkdir OUTPUT_FILES
$: mkdir OUTPUT_FILES/DATABASES_MPI
```

The input files for *SPECFEM3D* should be tested to make sure that they are ready for running any simulations. This means that the user should set up the meshing files at *DATA/meshfem3D_files*, initial xyz model at *DATA/tomo_files* or gll model in a user defined folder, and *DATA/Par_file*. Set `APPROXIMATE_HESS_KL = .true.` and `USE_RHO_SCALING = .true.` in *Par_file*, which will be used in **Stage II**.

For testing if the mesh and initial model is correctly set up, set `SAVE_MESH_FILE = .true.` in `Par_file` and plot the velocity model `procXXXXXX.bin` files in `OUTPUT_FILES/DATABASES_MPI`, and set it back to `.false.` after the test. Uncomment the following lines in the PBS script `pbs_fwatl_fwd_measure_adj.sh` or SLURM script `sbash_fwatl_fwd_measure_adj.sh`.

```
$: mpirun -np NPROC ./bin/xmeshfem3D
$: mpirun -np NPROC ./bin/xgenerate_databases
$: exit
```

The user should also check `output_mesher.txt` and `output_meshfem3D.txt` for the minimum period resolved, suggested time step, and other useful information regarding the meshing and solver parameters.

4.3 1.3 Prepare fwat_params

There are two user-defined parameter files of **FWAT**: `FWAT.PAR` and `MEASUREMENT.PAR`. The format of `FWAT.PAR` is:

```
##### Source #####
### Always set NSCOMP=1 for tele simulations and ANAT using only vertical components.
### NSCOMP>1 is not permitted in v1.1 version.
NSCOMP: 1
#
SCOMPS: Z
#
##### Receiver #####
NRCOMP: 1
# in order of Z R T
RCOMPS: Z
#
CH_CODE: BX
#
##### Filtering #####
NUM_FILTER: 3
#
SHORT_P: 6 10 15
#
LONG_P: 15 20 30
##### Measuring Window #####
### Noise FWI only, GROUPVEL_MIN and GROUPVEL_MAX are used to set time windows.
#
GROUPVEL_MIN: 2.65 2.65 2.65
#
GROUPVEL_MAX: 4.20 4.20 4.20
#
### TeleFWI only, TW_BEFORE and TW_AFTER are optional to set time windows
# (default is to read the time windows from FKtimes.dat) .
TW_BEFORE: 5
#
TW_AFTER: 45
# Set ADJ_SRC_NORM=.true. for noise FWI if normalizing adjoint sources at
# different period bands. Always set it .false. for other inversions.
ADJ_SRC_NORM: .true.
```

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```

# Noise FWI only, Set USE_NEAR_OFFSET=.false. to reject EGFs with distances
# less than one average wavelength
USE_NEAR_OFFSET: .true.
# Noise FWI only, Set SUPPRESS_EGF=.false. to convert noise correlations to EGFs.
# Set it .true. for checkerboard tests of noise FWI.
SUPPRESS_EGF: .false.
##### Postprocessing #####
# Set SAVE_OUTPUT_EACH_EVENT=.true. to save event kernels;
SAVE_OUTPUT_EACH_EVENT: .false.
# Horizontal radius of the Gaussian smoothing function, in meter
SIGMA_H: 10000
# Vertical radius of the Gaussian smoothing function, in meter
SIGMA_V: 10000
# Optimization method. Options: SD / CG / LBFGS
OPT_METHOD: SD
# Preconditioner. Options: default / z_precond
PRECOND_TYPE: default
# Maximum step length if not using a line search method
MAX_SLEN: 0.02
# Set DO_LS .true. if doing a line search
DO_LS: .true.
# Number of trial models for a line search
NUM_STEP: 5
# Step lengths of trial models for the line search. Format: F5.3
STEP_LENS: 0.010 0.020 0.030 0.040 0.050
#

```

The format of *MESUREMENT.PAR* is the same as that used in the software *meaure_adj*. An example is:

```

-2.0 0.025 6000 # tstart, DT, npts: time vector for simulations
              7 # imeas (1-8; see manual)
              BX # channel: BH or LH
30.000 6.000 # TLONG and TSHORT: band-pass periods for records
.false. # RUN_BANDPASS: use band-pass on records
.false. # DISPLAY_DETAILS
.false. # OUTPUT_MEASUREMENT_FILES
.true. # COMPUTE_ADJOINT_SOURCE
-4.5000 4.5000 # TSHIFT_MIN; TSHIFT_MAX
-1.5000 1.5000 # DLNA_MIN; DLNA_MAX
0.800 # CC_MIN
1 # ERROR_TYPE -- 0 none; 1 CC, MT-CC; 2 MT-jack-knife
1.000 # DT_SIGMA_MIN
0.500 # DLNA_SIGMA_MIN
1 # ITAPER -- taper type: 1 multi-taper; 2 cosine; 3 boxcar
0.020 2.50 # WTR, NPI (ntaper = 2*NPI)
2.000 # DT_FAC
2.500 # ERR_FAC
3.500 # DT_MAX_SCALE
1.500 # NCYCLE_IN_WINDOW
.false. # USE_PHYSICAL_DISPERSION

```

Note: Remember to change *tstart*, *DT*, *npts*, *imeas*, *channel*, *TSHIFT_MIN*/*TSHIFT_MAX*, *DLNA_MIN*/*DLNA_MAX*,

CC_MIN, ITAPER.

For FWAT of different data types (noise/tele), I recommend to use different parameter files, such as *FWAT.PAR.noise/FWAT.PAR.tele*, *MEASUREMENT.PAR.noise/MEASUREMENT.PAR.tele*.

4.4 1.4 Running forward and adjoint simulations (Stage I)

As long as the SEM mesh/model is set up, comment out the following lines in the PBS script *pbs_fwatl_fwd_measure_adj.sh* or SLURM script *sbash_fwatl_fwd_measure_adj.sh*.

```
#mpirun -np NPROC ./bin/xmeshfem3D
#mpirun -np NPROC ./bin/xgenerate_databases
#exit
```

and run the script *submit_job_fwatl.sh*. We should first set the model name (such as *mod=M01*), and step length (such as *step=0.02*) for reading updated model from previous iterations.

```
Usage: ./submit_job_fwatl.bash M?? simu_type setb sete
M??      --- name of the current model, such as M00, M01, ..., etc.
simu_type --- simulation type: noise, tele, or leq.
setb     --- begin number of sources_set, such as 1, 2, ..., etc.
sete     --- end number of sources_set, such as 1, 2, ..., etc.
```

This script will call *pbs_fwatl_fwd_measure_adj.sh* or *sbash_fwatl_fwd_measure_adj.sh* to submit a PBS/SLURM job for each event set, in which the MPI program *xfwat1_fwd_measure_adj* is called to run the forward and adjoint simulations.

```
USAGE: mpirun -np NPROC bin/xfwat1_fwd_measure_adj model set simu_type run_opt
model    --- name of current model, such as M00, M01, ...
set      --- name of event set, such as set1, set2, ...
simu_type --- simulation type: noise, tele, leq
run_opt  --- run_opt= 1 (fwd), 2 (fwd-meas), 3 (fwd-meas-adj)
```

Note the program has an option (*run_opt*) to run forward simulations only (*run_opt=1*), or with measuring misfit and adjoint sources (*run_opt=2*), and adjoint simulations (*run_opt=3*). The program can be divided into three parts:

1. Making forward simulation directory. The program will make all forward simulation directories under the directory **solver** (Figure 2) by looping over event sets defined by a variable *ipart* according to the file *sources_setXX.dat*.
2. Running *run_preprocessing.f90*. We adopt a multi-scale strategy in misfit measurement, thus the author should choose how many and what frequency bands needed in current iteration depending on the data, then make changes to *FWAT.PAR*. Note in file *run_preprocessing.f90*, we have three options (*simu_type*) to measure misfits and adjoint sources for data sets of noise, tele and leq. The misfits are stored in the directory **misfits** as shown in Figure 2.
3. Run adjoint simulations to obtain event kernels, sum and save them into one event gradient. In the *FWAT.PAR*, we have an option (*SAVE_OUTPUT_EACH_EVENT: .true.*) to save each event kernel.

4.5 1.5 Post-processing and model update (Stage II)

Then, run `submit_job_fwat2.sh` to do post-processing on event kernels and obtain the total misfit gradient and update the model.

```
Usage: ./submit_job_fwat2.bash M?? setb sete
M??      --- name of the current model, such as M00, M01, ..., etc.
setb     --- begin number of sources_set, such as 1, 2, ..., etc.
sete     --- end number of sources_set, such as 1, 2, ..., etc.
```

This script will call `pbs_fwat2_postproc_opt.sh` or `sbash_fwat2_postproc_opt.sh` to submit a PBS/SLURM job for a number of event sets, in which the MPI program `xfwat2_postproc_opt` is called to run postprocessing and optimization.

```
USAGE: mpirun -np NPROC bin/xfwat2_postproc_opt model setb sete is_smooth
model    --- name of current model, such as M00, M01, ...
setb     --- name of beginning event set, such as set1, set2, ...
sete     --- name of ending event set (sete>=setb), such as set1, set2, ...
is_smooth --- true or false, whether to do kernel smoothing or not.
```

This program can be divided into three parts:

- Summation and precondition of event kernels to obtain the final misfit gradient. The program will apply a preconditioner (PRECOND_TYPE: default/z_precond in *FWAT.PAR*) to each event kernel and sum all preconditioned event kernels to obtain the final misfit gradient in the directory `./optimize/SUM_KERNELS_M??` (Figure 2). Gradient files are in the format of `proc000??_alpha_kernel.bin`, `proc000??_beta_kernel.bin`, `proc000??_rho_kernel.bin`. I would recommend using PRECOND_TYPE: default for ANAT and PRECOND_TYPE: z_precond for TeleFWI.
- A 3-D Gaussian smoothing of the misfit gradient. Smoothed gradients are saved in the directory `./optimize/SUM_KERNELS_M??` with the format of `proc000??_alpha_kernel_smooth.bin`, `proc000??_beta_kernel_smooth.bin`, `proc000??_rho_kernel_smooth.bin`.
- Model update. There are two ways to update the model by either using a fixed step length (DO_LS: .false.) or an optimal step length based on a line search method (DO_LS: .true.).

Note: If you choose DO_LS: .false., then the model will be updated by a fix step length using MAX_SLEN. The output of the new model will be saved in the directory `./optimize/MODEL_M??`. Then, go back to **Stage I** and **II** to run the next iteration.

If you choose DO_LS: .true., the program will generate a number of trial models according the STEP_LENS such as `./optimize/MODEL_M??_step0.020`, `./optimize/MODEL_M??_step0.040`, etc. Then, you should run a line search in **Stage III**.

4.6 1.6 Line search for optimal step length (Stage III)

After all trial models are generated (DO_LS: .true.), we use them to do a line search in order to obtain the optimal step length. The script responsible for this is `submit_job_fwat3.sh`.

```
Usage: ./submit_job_fwat3.bash M?? simu_type
M??      --- name of the current model, such as M00, M01, ..., etc.
simu_type --- simulation type: noise, tele, or leq.
```

This script will call *pbs_fwat3_linesearch.sh* or *sbash_fwat3_linesearch.sh* to submit a PBS/SLURM job for each event set, in which the MPI program *xfwat3_linesearch* is called to run the forward simulations and misfit measuring.

```
USAGE: mpirun -np NPROC bin/xfwat3_linesearch model set simu_type
model    --- name of current model, such as M00_step0.020, M00_step0.040, ...
set      --- name of event set, set it to "ls".
simu_type --- simulation type: noise, tele, leq
```

The program will make all forward simulation directories by looping over step lengths, and each of them has a selected number of events from the file *sources_ls.dat*. The data processing is similar to *xfwat1_fwd_measure_adj* but with only forward simulations and misfit measuring.

As long as all the forward simulations for different trial models are finished, the corresponding traveltime misfits are collected in the directory **outputs/M??**. Then, we use the script *plt_line_search.mltiband.ANAT.bash* to plot the total misfit curve over step length and choose the optimal step length. After that, assign the trial model with the optimal step length (such as *./optimize/MODEL_M00_step0.040*) to our new model (*./optimize/MODEL_M01*) for running the next iteration.

CHANGELOG

5.1 FWAT v1.1

Released on Mar 24, 2021. Main contributors: Bin He, Kai Wang, Mijian Xu

- Update FWAT based on the latest stable version of *SPECFEM3D* released on 2018-08-30.
- Update FWAT.PAR.
- Compile with gfortran

5.2 FWAT v1.0

Released on Oct 6, 2020. Main contributor: Kai Wang

- FWAT of ambient noise , teleseismic P waves and their joint/alternating inversion. Codes tested in central California (Wang et al., 2020, JGR)
- First version of the inversion workflow based on *SPECFEM3D* released on 2018-02-11.

CONTRIBUTORS

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The **FWAT** package (v 1.0) was first developed by Kai Wang during his postdoc studies at Macquarie University (2020/2-present) and the University of Toronto (2018/12-2019/12). This inversion package has been tested across several seismic projects in research groups of Prof. Liu Qinya and Prof. Yingjie Yang.

Bin He, binhebj@gmail.com, joined in Mar 2021.

Modified files that interface with the *SPECFEM3D* package to update **FWAT** v1.0 to v1.1.

Mijian Xu, gomijianxu@gmail.com, joined in Jan 2022.

Working on **FWAT** of receiver functions, python tools for model visualization.

Nanqiao Du, nanqiao.du@mail.utoronto.ca, joined in Jan 2022.

Working on API interfaces.

Lurun Su, sulr1915935360@mail.ustc.edu.cn, joined in Apr 2023.

Working on **FWAT** examples.

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Working on **FWAT** examples.

Zengyu Luo, lbter@mail.ustc.edu.cn, joined in Nov 2023.

Working on **FWAT** of ZHratio.